

Render-to-display of analytic anti-aliased text and vector graphics in VR

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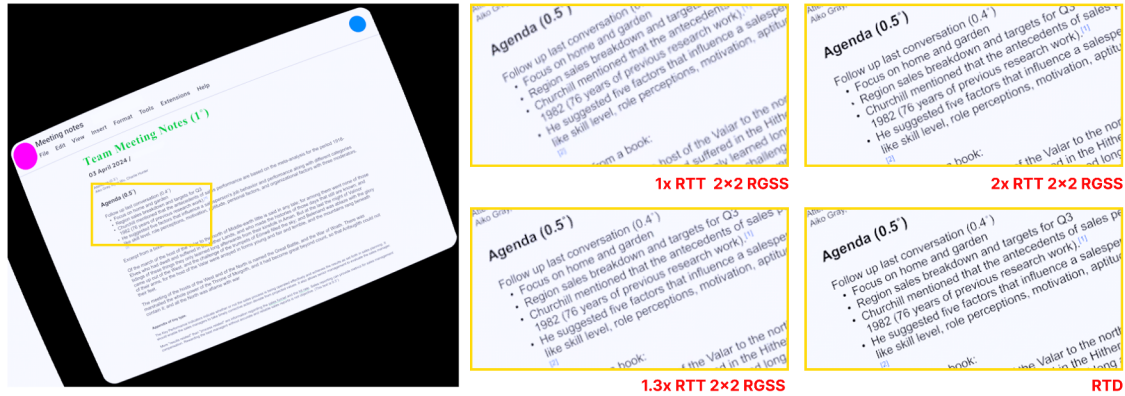


Figure 1: Render-to-display (RTD) Analytic anti-aliasing (AAA) produces higher quality text rendering compared to contemporary Render-to-texture (RTT) based techniques. Here RTD produces text that is more legible compared to text rendered at a resolution of various multiples of the display resolution and sampled using rotated-grid super sampling (RGSS).

ABSTRACT

Traditional render-to-texture (RTT) approaches struggle to deliver high-fidelity text in VR environments across various scene and display configurations. Analytic anti-aliasing (AAA) achieves super-sampled quality at significantly lower computational cost, yet its integration into real-time VR systems remains unexplored. We present an end-to-end render-to-display (RTD) pipeline incorporating AAA for text and vector graphics in VR. Leveraging the Skia 2D library, our system extracts glyph data to analytically compute pixel coverage. Our novel render pipeline for VR demonstrates superior text quality compared to RTT methods, validated through user studies highlighting perceptual improvements.

Index Terms: Text and Vector Graphics Rendering, Anti-aliasing, Virtual Reality, Augmented Reality.

1 DESCRIPTION AND METHODOLOGY

In modern VR systems, the background 3D layer and foreground 2D UI layers are rendered separately before being composited together [2]. This separation allows for independent layer resolutions and prevents double sampling (see Fig. 2). However, sampling and filtering are not eliminated from the text rendering pipeline completely.

RTD techniques for rendering text have been demonstrated on desktop systems to achieve 16x SSAA equivalent quality by analytically computing pixel coverage directly from vector shapes (Hunt et al. [1]), while delivering orders-of-magnitude better performance by eliminating intermediate formats, resampling, and filtering.

In order to study the feasibility of adoption of RTD AAA in VR, we have modified a popular high-value mobile content source that provides the 2D curve data and also developed a custom render

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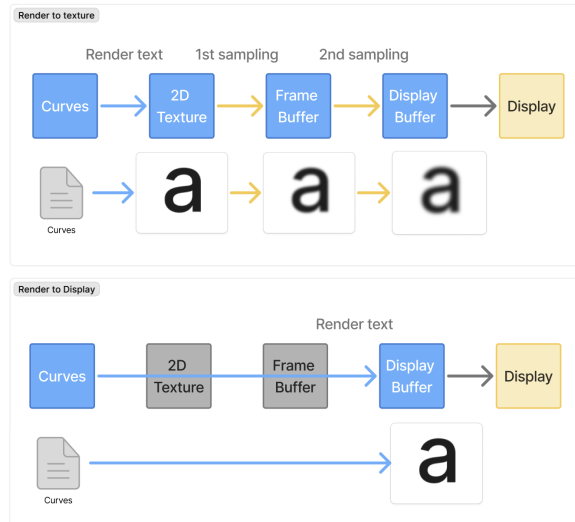


Figure 2: Traditional RTT pipeline suffers from double sampling unlike the RTD pipeline.

pipeline that computes pixel coverage of these curves in distorted space. We then perform user studies comparing RTT and RTD to study the trade-offs using this system.

1.1 Content pipeline

Our content pipeline is built on the Skia 2D rendering library, which is widely used. The pipeline includes several additions to Skia, and the Chromium engine has been modified to work with this new Skia interface. The high-level Skia canvas object is backed by a lower-level device object. Changes to Chromium involves duplicating

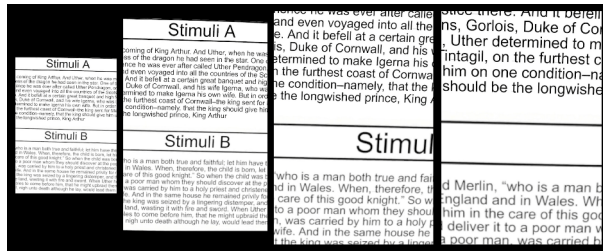


Figure 3: Text panel in VR rendered with RTD AAA (top) and RTT (bottom) at various distances. RTD is resolution independent.

render commands to a second Skia canvas backed by our custom Skia device. We provide a new device implementation, which leverages Skia’s various routines for processing and simplifying vector graphics to collapse high-level concepts down to simple paths (line and quadratic segments). Once simplified, Skia path data is sent over a pipe to the render process using a set of simple message structs or written to disk.

1.2 Rendering pipeline

The scene data received from Skia consists of paths. A driver compiles the path data into a “vector graphics texture”. This texture is a tiled data structure consisting of variable-sized command lists in each tile. Each tile that a path intersects is filled with appropriate commands and other data such as background color, fill rules, gradients, bitmaps and so on. During compilation the paths are decomposed into simpler trapezoids and quadratic curves. A GPU shader or ideally a GPU HW sampler can then compute pixel coverage by looking up the shapes that occur at that pixel and computing coverage area of trapezoids and quadratic curves [1].

2 RESULTS

Compared with state-of-the-art RTT algorithms used in contemporary VR (“Quality Supersampling” [3]), RTD AAA produces text that looks most legible across various scene configurations. (see Fig. 1). Additionally RTD AAA is texture resolution independent so it does not suffer from quality degradation on magnification that RTT techniques suffer from (see Fig. 3), so no application re-render is necessary.

2.1 VR user study

We performed a within-subject user study with 44 attendees. The same text content was rendered using RTT and RTD for resolutions of 40 and 60 pixels-per-degree (PPD), simulated on a 120 PPD prototype VR headset. The study was designed as a blind A/B test, with participants not aware whether they were evaluating RTD or RTT. In all scenarios, participants indicated their preference for the left (RTD) or right (RTT) method using a 5-point Likert scale. This scale ranged from 1 (strong preference for left) to 5 (strong preference for right), with 3 indicating an indistinguishable preference. This study offers two main observations (see Fig. 4). Initially, when both the RTD and RTT methods used the same PPD, participants significantly favored the RTD method ($p < 0.0001$, with each response considered equally likely). This indicates that users may perceive RTD as superior or preferable at the same PPD (Fig. 5). Furthermore, when lower PPDs in RTD were tested against higher PPDs in RTT, there was no significant preference among participants for either method. This lack of preference suggests that even at lower PPDs, RTD’s perceived quality can compete with RTT’s, despite RTT employing higher PPDs. As a result, choosing lower PPD displays with RTD could be a viable option for future VR products without noticeably affecting the user experience.

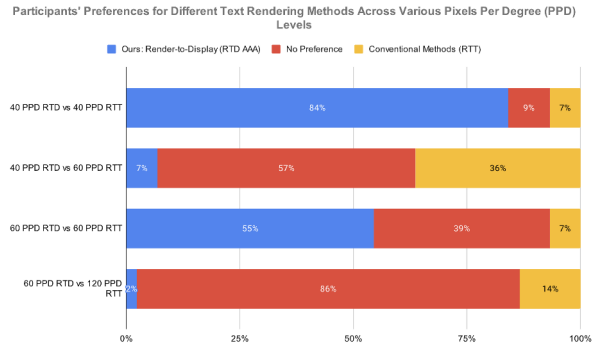


Figure 4: User study results comparing RTD and RTT at various PPDs

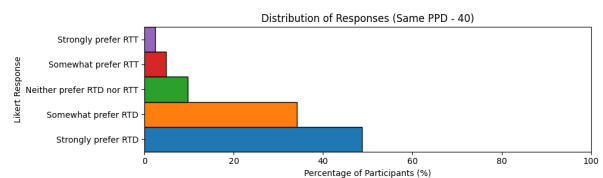


Figure 5: Comparing RTD and RTT at the same PPD (40). Strong preference is seen for RTD ($p < 0.0001$).

This could lead to lower costs and more available GPU compute resources for other uses. The same result was seen on a similar study with lower PPDs of 24 and 30. (Note: Informed consent was obtained from all participants prior to participation in the user study.)

2.2 Open source

The content pipeline based on Chromium and Skia, a reference driver, GPU shader based vector graphics renderer, a fixed-point functional model and the associated tool chain are available as open source. Please see supplemental materials for details.

SUPPLEMENTAL MATERIALS

All supplemental materials are available on GitHub at <https://github.com/facebookresearch/AnalyticTextXR>, released under a CC BY-NC 4.0 license.

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